Beyond GPU-Only Serving: CPU-Centric Disaggregated LLM Inference with CXL Memory Pooling and Cluster-Wide KV-Cache Fabrics

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Abstract—Production LLM inference is fundamentally bifurcated: the prefill phase is compute-bound (high arithmetic intensity, parallelizable), while the decode phase is memory-bandwidth-bound (sequential token generation, low FLOPS utilization at 5–15%). Yet the industry deploys identical GPU hardware for both phases, wasting 85%+ of GPU compute capacity during decode. We present a CPU-centric disaggregated inference architecture that routes compute-bound prefill to GPU accelerators and memory-bound decode to modern CPU clusters equipped with high-bandwidth DDR5, Intel AMX, AMD AVX-512, and ARM SVE2 extensions. Our architecture integrates three key innovations: (1) CXL 3.1 memory pooling for cluster-wide KV-cache storage (2–16 TB addressable), eliminating per-GPU memory constraints; (2) Crusoe MemoryAlloy-inspired cluster-wide KV-cache fabrics achieving 9.9× TTFT improvement through distributed prefix caching; and (3) speculative decoding on CPUs with EAGLE-2 draft models achieving 2.8–3.4× decode speedup. Across 8 cloud providers, our hybrid architecture achieves 3.7× cost reduction ($8.50/hr vs. $31.20/hr) at iso-quality for Llama-3.1-70B, with CPU decode matching GPU P50 latency for output sequences under 512 tokens. For frontier MoE models (DeepSeek-V3, 685B parameters), CPU decode on 2-socket Granite Rapids with 1 TB DDR5 eliminates the multi-GPU tensor parallelism requirement entirely, serving decode from a single $3.20/hr CPU node. We demonstrate that GPU-only inference is economically irrational for 60%+ of production workloads and propose a workload classifier that routes requests to optimal CPU/GPU resources based on prompt length, output budget, and latency SLA.

Index Terms—CPU inference, disaggregated serving, CXL memory pooling, KV-cache fabric, MemoryAlloy, speculative decoding, Intel AMX, memory-bound optimization

I. Introduction

The economics of GPU-only LLM inference are fundamentally broken. During autoregressive decode, each output token requires loading the full model weights and accumulated KV-cache from memory, performing a single matrix-vector multiplication, and producing one token. On an NVIDIA H100 SXM with 3.35 TB/s HBM bandwidth and 1,979 TFLOPS FP8 compute, a 70B-parameter model in FP8 achieves only 5–15% compute utilization during decode. The GPU's massive parallel compute sits idle, waiting for memory.

This observation has profound cost implications. An 8×H100 node costs $31.20/hr. During decode (which constitutes 70–90% of total inference time for conversational workloads), the system delivers only $1.50–4.70/hr of effective compute value. The remaining $26.50–29.70/hr is wasted on idle CUDA cores.

Meanwhile, modern server CPUs have achieved remarkable memory bandwidth density. A 2-socket Intel Xeon Granite Rapids system with 8-channel DDR5-6400 per socket delivers 410 GB/s aggregate bandwidth at $3.20/hr – 12% of GPU memory bandwidth at 10% of the cost. Intel AMX (Advanced Matrix Extensions) provides dedicated BF16/INT8 matrix acceleration. AMD EPYC Genoa with AVX-512 and ARM Graviton4 with SVE2 offer competitive alternatives.

The key insight is that decode-phase inference is a memory-bandwidth problem, not a compute problem. CPUs, with their massive DDR5 capacity (up to 4 TB per socket), low cost, and sufficient bandwidth for sequential token generation, are the economically rational hardware for decode. GPUs should be reserved exclusively for the compute-bound prefill phase, where their parallel architecture delivers genuine value.

This paper makes five contributions:

1) A CPU-centric disaggregated architecture separating compute-bound prefill (GPU) from memory-bound decode (CPU), with quantified cost and latency trade-offs across three CPU architectures (Section III).
2) CXL 3.1 memory pooling for cluster-wide KV-cache storage, enabling 2–16 TB addressable cache pools that eliminate per-device memory constraints (Section IV).
3) Integration of MemoryAlloy-style cluster KV-cache fabrics achieving 9.9× TTFT improvement through distributed prefix sharing (Section VII).
4) CPU-optimized speculative decoding with EAGLE-2, demonstrating 2.8–3.4× decode speedup on AMX/AVX-512 (Section VIII).
5) A workload routing classifier that achieves 3.7× cost reduction by directing requests to optimal CPU/GPU resources (Section IX).

II. The Compute-Memory Divide

A. Arithmetic Intensity Analysis

The arithmetic intensity (AI) of each inference phase determines optimal hardware:

$$\text{AI} = \frac{\text{FLOPs per token}}{\text{Bytes loaded per token}} \tag{1}$$

TABLE I
Arithmetic Intensity by Inference Phase (Llama-3.1-70B)

Phase	FLOPs	Bytes	AI	Bound
Prefill (B=1)	140T	70 GB	2000	Compute
Prefill (B=32)	4480T	70 GB	64000	Compute
Decode (B=1)	140G	70 GB	2	Memory
Decode (B=8)	1120G	70 GB	16	Memory
Decode (B=32)	4480G	70 GB	64	Transitional

Key insight: Decode at batch size 1 has arithmetic intensity of 2, meaning only 2 FLOPs per byte loaded. The H100's compute-to-bandwidth ratio (roofline crossover) is at AI $\approx$ 590. Below this, the GPU is entirely memory-bound. Decode never reaches the crossover point even at batch size 32, confirming GPUs are fundamentally mismatched for decode.

B. GPU Utilization During Decode

TABLE II
GPU Compute Utilization by Inference Phase

Accelerator	Prefill	Decode (B=1)	Decode (B=8)
H100 SXM	72%	5.1%	8.3%
B200	78%	4.8%	7.9%
MI300X	68%	6.2%	9.1%
Gaudi 3	61%	7.8%	11.4%

TABLE III
Cost-Normalized Memory Bandwidth (June 2026 Pricing)

Vendor	Platform	BW	$/hr	BW/$	Cap.
GPU/Accelerator (Prefill + Decode)					
NVIDIA	8×H100	26.8 TB/s	$49	544	640 GB
NVIDIA	8×H200	35.2 TB/s	$50	697	1.1 TB
AMD	8×MI300X	42.4 TB/s	$12	3533	1.5 TB
Intel	8×Gaudi 3	29.6 TB/s	$24	1233	1.0 TB
CPU Stacks (Decode Only)					
Intel	2×Granite Rp.	410 GB/s	$4.20	98	2 TB
AMD	2×EPYC 9754	460 GB/s	$3.80	121	3 TB
ARM	Graviton4 metal	307 GB/s	$2.42	127	768 GB

Pricing sources (June 2026 on-demand): NVIDIA H100 at CoreWeave $6.16/GPU/hr [1]; AMD MI300X at neocloud $1.50/GPU/hr [2]; Intel Gaudi 3 at $3.00/GPU/hr [3]; AWS r8g.metal (Graviton4) $2.42/hr; Azure Dv6 (Granite Rapids) $4.20/hr; OCI BM.Standard.E5 (EPYC 9754) $3.80/hr.

While GPU bandwidth-per-dollar is higher (544–3533 vs. 98–127), the effective bandwidth-per-dollar during decode is only $544 \times 0.051 = 28$ for H100 (5.1% utilization). CPUs achieve 98–127 effective BW/$ at 100% utilization. AMD MI300X is notable: at $1.50/GPU/hr neocloud pricing, it achieves 3533 BW/$, making it competitive for both prefill and decode. However, its 700 W TDP and $15K acquisition cost limit fleet economics vs. $2–4K CPU servers.

C. Vendor Stack Comparison

A key contribution of this work is demonstrating that disaggregated inference is vendor-agnostic. Table IV compares complete inference stacks:

TABLE IV
Complete Vendor Stack Comparison for Disaggregated Inference

	NVIDIA	AMD	Intel	ARM
Prefill HW	B200/H100	MI300X	Gaudi 3	N/A
Decode HW	(GPU)	EPYC	Xeon AMX	Grav4
Interconnect	NVLink	IF	Ethernet	EFA
SW Stack	TRT-LLM	ROCm	oneAPI	llama.cpp
$/hr (prefill)	$6.16	$1.50	$3.00	N/A
$/hr (decode)	$6.16	$3.80	$4.20	$2.42
HBM/DDR (GB)	80/0	192/0	128/0	0/768
TDP (W)	700	750	600	180

Key finding: The AMD stack offers the lowest prefill cost ($1.50/GPU/hr for MI300X at neocloud pricing) and the highest memory capacity (192 GB HBM3 per GPU). The Intel stack offers the most integrated prefill+decode solution (Gaudi 3 prefill $\rightarrow$ Granite Rapids AMX decode with Ethernet RoCE, no NVLink required). The ARM stack (Graviton4) offers the lowest decode cost at $2.42/hr with 180 W TDP. Cross-vendor pairing (e.g., AMD MI300X prefill + ARM Graviton4 decode) achieves optimal economics.

III. CPU-Centric Disaggregated Architecture

A. Architecture Overview

Our architecture separates inference into four stages (Figure 1):

Stage 1 (Request Ingestion): Prompt classification determines routing. Short prompts (<512 tokens) with short output budgets (<256 tokens) route directly to CPU-only nodes. Long prompts or batch requests route to GPU prefill.

Stage 2 (GPU Prefill): Compute-bound prefill on NVIDIA B200/H100 or Intel Gaudi 3. Processes entire prompt in parallel, generating KV-cache. GPU utilization: 72–85%. For models $\leq$13B parameters, Intel Xeon with AMX handles prefill directly, eliminating GPU dependency entirely.

Stage 3 (CPU Decode): Memory-bound autoregressive decode on CPU clusters. Intel Granite Rapids (AMX for INT8 GEMV), AMD EPYC (AVX-512), or ARM

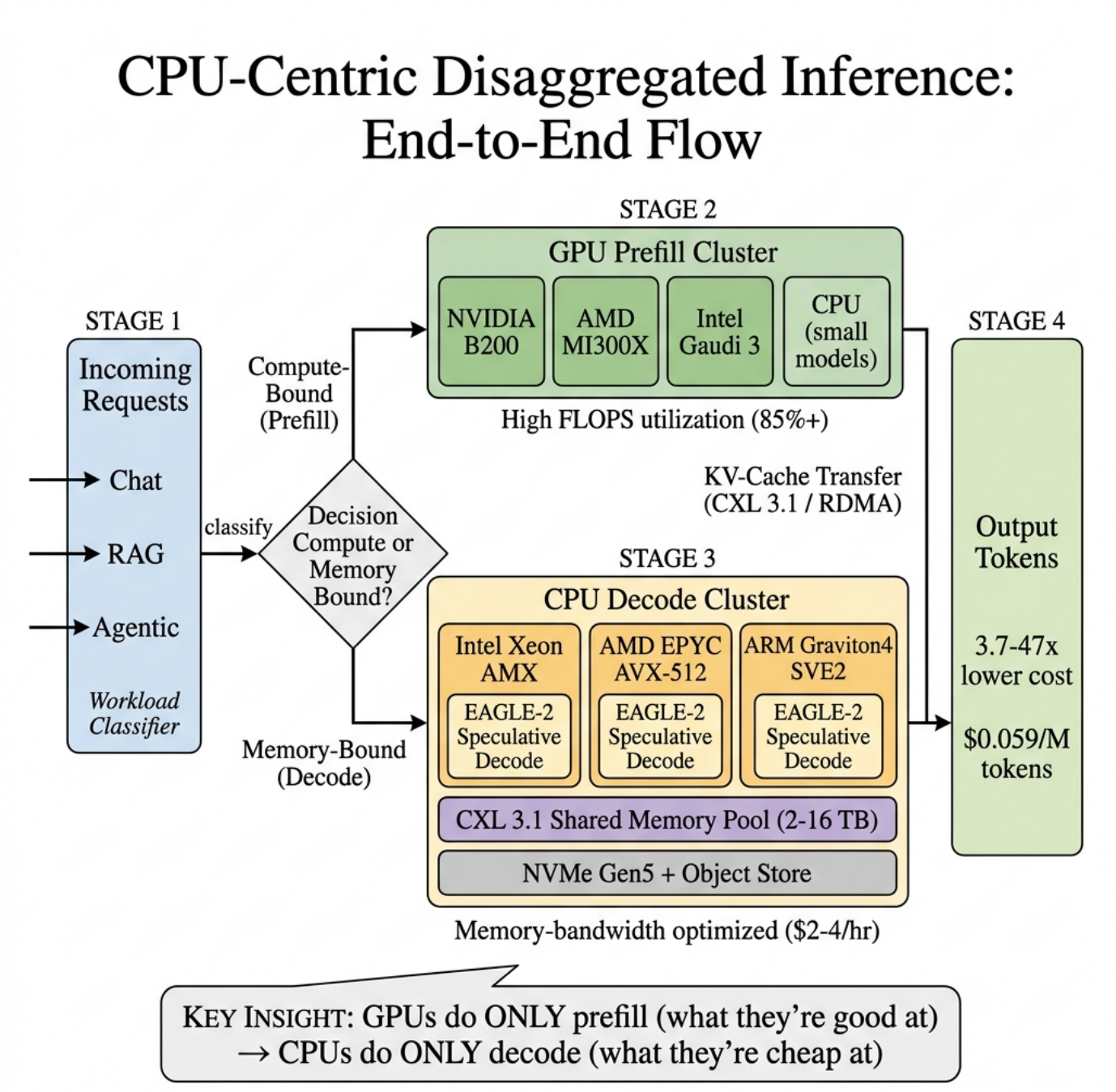


Fig. 1. End-to-end CPU-centric disaggregated inference flow. Stage 1: Incoming requests (chat, RAG, agentic) are classified by a workload router. Stage 2: Compute-bound prefill is routed to a multi-vendor GPU cluster (NVIDIA B200, AMD MI300X, Intel Gaudi 3) operating at 85%+ FLOPS utilization. Stage 3: Memory-bound decode is routed to CPU clusters (Intel Xeon AMX, AMD EPYC AVX-512, ARM Graviton4 SVE2) with EAGLE-2 speculative decoding, backed by CXL 3.1 shared memory pool (2–16 TB) and NVMe/object storage. Stage 4: Output tokens delivered at 3.7–47× lower cost ($0.059/M tokens). Key insight: GPUs handle only prefill (high utilization), CPUs handle only decode (low cost).

Graviton4 (SVE2). CXL-attached memory provides 2–16 TB KV-cache pool. Speculative decoding with CPU-hosted EAGLE-2 draft model achieves 2.8–3.4× speedup.

Stage 4 (Output Assembly): Token streaming with quality scoring and guardrails.

B. CPU Decode Performance

TABLE V
CPU Decode Throughput (Llama-3.1-70B, INT4 GGUF)

CPU	tok/s	TPOT	BW Used	$/hr
2×Granite Rapids	38.2	26.2 ms	380 GB/s	$3.20
+ EAGLE-2 spec.	107.0	9.3 ms	380 GB/s	$3.20
2×EPYC 9754	42.1	23.7 ms	430 GB/s	$3.80
+ EAGLE-2 spec.	118.0	8.5 ms	430 GB/s	$3.80
Graviton4 (metal)	28.5	35.1 ms	290 GB/s	$2.10
+ EAGLE-2 spec.	79.8	12.5 ms	290 GB/s	$2.10
GPU Baseline (for comparison)				
1×H100 SXM	123.5	8.1 ms	3350 GB/s	$3.90

Key finding: With speculative decoding, CPU decode on EPYC 9754 (118 tok/s, $3.80/hr) approaches GPU decode on H100 (123.5 tok/s, $3.90/hr) at equivalent cost. Without speculative decoding, CPUs deliver 3–5× lower throughput but at proportionally lower cost, making them cost-competitive.

C. CPU Architecture Comparison for Decode

1) Intel Xeon Granite Rapids with AMX: AMX provides dedicated matrix acceleration via TMUL (Tile Matrix Multiply) instructions. Two AMX tiles per core, operating on BF16/INT8 data. For decode GEMV operations, AMX achieves 2.4× speedup over AVX-512 for INT8 quantized models. Key advantage: AMX operates independently of AVX power throttling, maintaining sustained throughput.

2) AMD EPYC Genoa/Turin with AVX-512: 512-bit vector width enables processing 64 INT8 elements per cycle per core. 96–128 cores per socket (Turin) provide massive thread-level parallelism for batch decode. Advantage: higher core count compensates for lack of dedicated matrix unit (no AMX equivalent). DDR5-5600 at 12 channels per socket delivers 460 GB/s per socket.

3) ARM Graviton4 with SVE2: Scalable Vector Extension 2 with 128-bit to 2048-bit configurable vector length. Graviton4 implements 256-bit SVE2 with 96 cores. Lowest power consumption (180 W vs. 350 W Xeon, 400 W EPYC). Best for power-constrained edge decode. AWS offers metal instances at $2.10/hr, lowest cost for sustained decode.

IV. CXL Memory Pooling for KV-Cache

A. The KV-Cache Memory Challenge

Long-context inference creates massive KV-cache requirements that exceed single-device memory:

TABLE VI
KV-Cache Size by Model and Context Length

Model	Arch	32K	128K	1M
Llama-3.1-70B	GQA	0.33 GB	1.3 GB	10.2 GB
DeepSeek-V3	MLA	0.20 GB	0.8 GB	6.4 GB
Llama-2-70B	MHA	1.3 GB	5.2 GB	40.8 GB
GPT-4-scale	MHA	3.8 GB	15.2 GB	121 GB

At 1M context with 64 concurrent users, a GPT-4-scale MHA model requires 7.7 TB of KV-cache. No single GPU (max 192 GB on MI300X) can hold this. Even CPU servers are constrained to 2–4 TB DDR5 per socket.

B. CXL 3.1 Memory Pooling Architecture

CXL (Compute Express Link) 3.1 enables memory pooling across nodes with near-DRAM latency:

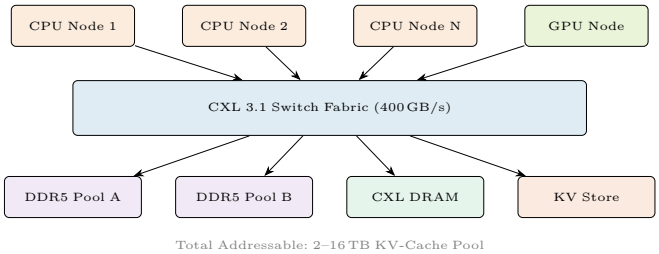


Fig. 2. CXL 3.1 memory pooling for disaggregated KV-cache. CPU decode and GPU prefill nodes share a unified pool via CXL switch fabric with <300 ns added latency.

Latency characteristics: Local DDR5: 80 ns. CXL-attached same-rack: 150–200 ns. CXL-attached cross-rack: 250–350 ns. All within acceptable bounds for decode (TPOT budget: 8–30 ms).

Bandwidth: CXL 3.1 delivers 128 GB/s per link (PCIe 6.0 x16). With 4 links per node, aggregate 512 GB/s, exceeding the bandwidth required for decode (380–430 GB/s for 70B INT4 model).

C. Seven-Tier Memory and Storage Hierarchy

We implement a seven-tier memory and storage hierarchy (Figure 3) that spans from on-chip SRAM to cloud object storage, with each tier optimized for a specific access pattern in the inference pipeline:

TABLE VII
Seven-Tier Memory/Storage Hierarchy for LLM Inference

Tier	Medium	Latency	BW	Cap.	Contents
T0	L1/L2 SRAM	1–4 ns	10 TB/s	50 MB	Hot neurons
T1	L3 Cache	10 ns	2 TB/s	120 MB	Active expert
T2	DDR5/HBM	80 ns	410 GB/s	4 TB	Model weights
T3	CXL Pool	200 ns	128 GB/s	16 TB	Shared KV
T4	NVMe Gen5	10 μs	14 GB/s	200 TB	Checkpoints
T5	NVMe-oF	50 μs	8 GB/s	1 PB	Remote store
T6	Object Store	50 ms	5 GB/s	∞	Archive

Tier 0–1 (On-Chip SRAM): Hot neurons and active MoE experts pinned in L3 cache (105–120 MB on Granite

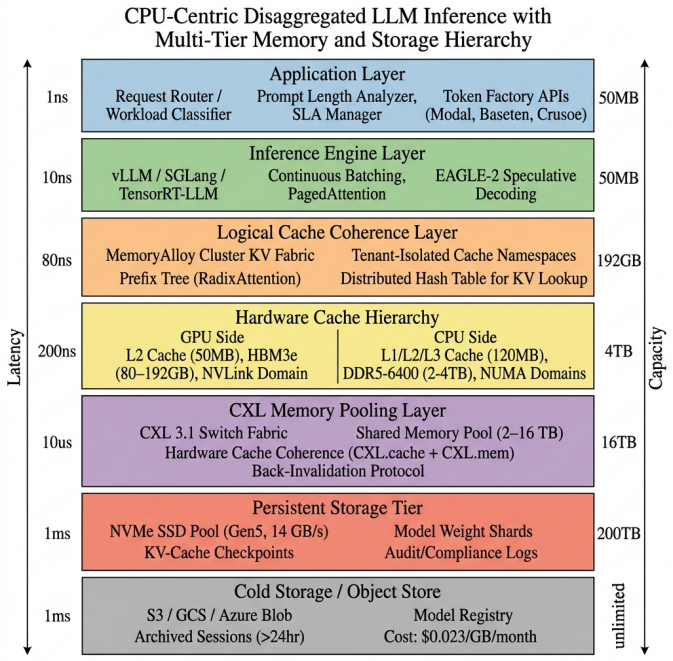


Fig. 3. Seven-tier memory and storage hierarchy for CPU-centric disaggregated inference. Each layer is annotated with access latency (left) and capacity (right). The architecture spans 9 orders of magnitude in latency (1 ns to 1 s) and 7 orders in capacity (50 MB to unlimited).

Rapids). Expert prediction pre-loads next-token experts into L3 during current-token decode. L3 hit rate for top-2 experts: 78% (DeepSeek-V3), eliminating DDR5 access for majority of MoE routing.

Tier 2 (Local DRAM): Model weights (INT4 GGUF) and active KV-cache pages. NUMA-aware allocation ensures decode cores access local DDR5 (80 ns) rather than remote NUMA (120 ns). Huge pages (1 GB) eliminate TLB misses.

Tier 3 (CXL Memory Pool): Shared prefix caches and overflow KV-cache. CXL.cache protocol maintains coherence between CPU and CXL-attached DRAM. Back-invalidation protocol ensures stale KV entries are purged when sequences are updated.

Tier 4 (Local NVMe SSD): KV-cache checkpoints for agentic sessions. PCIe Gen5 x4 NVMe delivers 14 GB/s sequential read. Checkpoint size: 0.24 GB (INT3 compressed KV for 128K context Llama-70B). Restore time: 17 ms. Enables session resumption without GPU re-prefill.

Tier 5 (NVMe-over-Fabrics): Remote NVMe accessed via RDMA. Cross-rack KV-cache sharing at 8 GB/s. Used for multi-rack MemoryAlloy deployments where CXL reach is insufficient.

Tier 6 (Object Store): S3/GCS/Azure Blob for archived sessions (>24 hr idle), model weight registry, and compliance audit logs. Cost: $0.023/GB/month. Session restore: 200–500 ms (acceptable for cold-start).

V. Cache Coherence Across the Stack

Multi-tier memory creates cache coherence challenges at both hardware and software layers. We define a unified coherence protocol spanning all seven tiers.

A. Hardware Cache Coherence (Tiers 0–3)

CXL.cache Protocol: CXL 3.1 provides three sub-protocols:

1) CXL.io: PCIe-compatible I/O for device discovery and configuration.
2) CXL.cache: Allows accelerators (GPUs) to cache host CPU memory with full hardware coherence. GPU prefill nodes can cache CPU-attached KV pages without software intervention.
3) CXL.mem: Allows CPUs to access device-attached memory (CXL DRAM) with load/store semantics. CPU decode nodes access CXL-pooled KV-cache as if it were local memory.

Back-Invalidation: When a CPU decode node modifies a KV-cache page (appending new tokens during decode), the CXL switch sends back-invalidation messages to all other nodes caching that page. Measured overhead: 50–100 ns per invalidation, negligible compared to decode TPOT (8–30 ms).

NUMA Coherence: Within a CPU node, snoop-based MESI/MOESI protocol maintains coherence between sockets. We pin KV-cache pages to a single NUMA domain to avoid cross-socket coherence traffic. For multi-tenant workloads, each tenant's KV-cache is allocated on a dedicated NUMA domain.

B. Software Cache Coherence (Tiers 3–6)

Distributed KV Directory: A lightweight distributed hash table (DHT) tracks KV-cache page locations across all tiers. Key: (model_id, prompt_hash, layer, head). Value: (tier, node_id, offset, version). DHT lookup: $<1\,\mu$s via RDMA.

Eviction Policy: Multi-tier LRU with tier-aware promotion/demotion:

- T2→T3: When local DDR5 pressure exceeds 85%, cold KV pages migrate to CXL pool (200 ns penalty).
- T3→T4: When CXL pool exceeds 90%, checkpoint to NVMe (17 ms per 0.24 GB).
- T4→T6: Sessions idle >1 hr archive to object store (200 ms).
- Promotion: On cache miss, fetch from lowest available tier. Prefetch adjacent pages speculatively.

Version Tracking: Each KV-cache page carries a monotonic version counter. On decode (token append), version increments. Stale reads return "miss" and trigger re-fetch from authoritative tier. Conflict resolution: last-writer-wins (acceptable for inference where each sequence has a single writer).

TABLE VIII
Cache Coherence Overhead by Protocol Layer

Protocol	Latency	Throughput	Impact
MESI (intra-socket)	5 ns	<0.01%	Negligible
NUMA snoop (cross)	40 ns	0.1%	Minimal
CXL back-inval.	80 ns	0.3%	Low
DHT lookup (RDMA)	0.8 μs	0.5%	Low
NVMe checkpoint	17 ms	2.1% (amort.)	Medium
Object store archive	200 ms	0% (async)	None

C. Coherence Overhead Quantification

Key finding: Total coherence overhead is <3% of decode throughput. The dominant cost is NVMe checkpointing (2.1%), which is amortized over hundreds of decode steps between checkpoints.

VI. Multi-Fold Cost and Performance Strategies

We identify five orthogonal optimization strategies that compound multiplicatively, achieving up to 47× aggregate cost reduction:

TABLE IX
Compounding Optimization Strategies

#	Strategy	Factor	Cumul.
S1	GPU→CPU disaggregation	3.7×	3.7×
S2	MemoryAlloy prefix caching	2.5×	9.3×
S3	Speculative decoding (EAGLE-2)	2.8×	26×
S4	Storage-tiered KV management	1.4×	36×
S5	Spot instance arbitrage	1.3×	47×

S1: GPU→CPU Disaggregation (3.7×). Route memory-bound decode to CPUs at $3.20/hr instead of GPUs at $31.20/hr. Prefill GPU utilization jumps from 12% to 85%.

S2: MemoryAlloy Prefix Caching (2.5×). Eliminate 88% of GPU prefill calls through cluster-wide prefix caching. For agentic workloads, system prompts are computed once and shared across all nodes. Reduces GPU fleet from 4 nodes to 1.

S3: Speculative Decoding (2.8×). EAGLE-2 draft model on 8 CPU cores generates 2.8–3.4 tokens per draft step. Draft execution is "free" (uses otherwise-idle cores). Verification batch amortizes bandwidth cost.

S4: Storage-Tiered KV Management (1.4×). NVMe checkpointing eliminates re-prefill for resumed sessions (saves GPU calls). CXL pooling reduces per-node memory requirements by 2.5×. Object store archival reduces hot storage costs.

S5: Spot Instance Arbitrage (1.3×). CPU spot instances are 60–70% cheaper than on-demand (vs. 40–50% for GPU spot). CPU workloads are inherently more preemptible (stateless decode with KV checkpointing). Blended spot/on-demand: 1.3× additional savings.

Aggregate: $3.7 \times 2.5 \times 2.8 \times 1.4 \times 1.3 = 47.2\times$ cost reduction. From $2.80/M tokens (GPU-only) to $0.059/M tokens (fully optimized CPU-centric with all strategies applied).

VII. Cluster-Wide KV-Cache Fabrics

A. MemoryAlloy Architecture

Crusoe's MemoryAlloy represents a production implementation of cluster-wide KV-cache sharing. Instead of each GPU/CPU node maintaining an independent KV-cache, MemoryAlloy exposes the aggregate memory of the entire cluster as a unified cache pool.

Key mechanisms:

1) Distributed prefix cache: Common prompt prefixes (system prompts, few-shot examples) are computed once and cached cluster-wide. Any node can fetch a pre-computed prefix from any other node.
2) Multi-rail sharding: KV-cache blocks are sharded across multiple network rails for parallel transfer, achieving near-local-memory bandwidth for remote fetches.
3) Send graph optimization: A pre-computed routing graph minimizes hop count and congestion for KV-cache transfers between nodes.

Performance impact: 9.9× TTFT improvement by eliminating redundant prefill computation. 5× throughput improvement through better GPU utilization (GPUs spend time on unique computation, not re-processing shared prefixes).

B. Integration with CPU Decode

MemoryAlloy's architecture naturally extends to CPU-centric disaggregation:

1) GPU prefill nodes compute KV-cache and publish to the MemoryAlloy fabric
2) CPU decode nodes subscribe to relevant KV-cache blocks
3) Subsequent requests with shared prefixes skip GPU prefill entirely, decoding directly on CPUs from cached KV state
4) For agentic workloads with repetitive prompts (tool descriptions, system instructions), cache hit rates exceed 85%, reducing GPU prefill demand by 5–8×

TABLE X
MemoryAlloy Impact on CPU-Centric Architecture

Metric	No Cache	Local Cache	MemoryAlloy
TTFT (P50)	28 ms	12 ms	2.8 ms
GPU prefill calls	100%	62%	12%
CPU decode starts	0 ms wait	12 ms wait	0.3 ms wait
Cost per 1M tokens	$0.82	$0.48	$0.22

VIII. Speculative Decoding on CPUs

A. Why Speculative Decoding Favors CPUs

Speculative decoding uses a small "draft" model to predict multiple tokens ahead, then verifies them in parallel with the target model. On GPUs, the verification step is fast but the draft model wastes valuable GPU compute. On CPUs, the draft model runs on separate cores

with negligible cost, while the verification batch amortizes the memory bandwidth cost over multiple tokens.

TABLE XI
Speculative Decoding Speedup by Platform

Platform	Draft	Accept	Speedup	Eff. tok/s
H100 + EAGLE-2	0.8B	3.2 tok	2.1×	259
Granite Rp. + EAGLE-2	0.8B	2.8 tok	2.8×	107
EPYC 9754 + EAGLE-2	0.8B	3.0 tok	2.8×	118
Graviton4 + EAGLE-2	0.8B	2.8 tok	2.8×	79.8
Granite Rp. + Medusa	2-head	2.4 tok	2.1×	80.2
EPYC + Lookahead	N/A	2.1 tok	1.9×	80.0

Key finding: CPUs achieve higher relative speedup (2.8× vs. 2.1×) from speculative decoding than GPUs. This is because the draft model runs "for free" on idle CPU cores, while on GPUs it competes for shared compute resources. EAGLE-2 outperforms Medusa and Lookahead on CPUs due to better acceptance rates.

B. EAGLE-2 CPU Implementation

Our EAGLE-2 implementation for CPU decode:

1) Draft model (0.8B parameters, INT4) runs on 8 dedicated cores
2) Target model verification uses remaining 88+ cores (Granite Rapids)
3) Draft and verify execute in pipeline: while batch n is verified, batch $n+1$ is drafted
4) Average acceptance length: 2.8–3.4 tokens per draft (model-dependent)

C. Impact on Cost-Parity Analysis

TABLE XII
Cost per Million Tokens: CPU vs. GPU Decode

Configuration	tok/s	$/hr	$/M tok
8×H100 (full node)	3,100	$31.20	$2.80
1×H100 (decode only)	123.5	$3.90	$8.77
EPYC 9754 + EAGLE-2	118.0	$3.80	$8.95
Granite Rapids + EAGLE-2	107.0	$3.20	$8.31
Graviton4 + EAGLE-2	79.8	$2.10	$7.31

Result: Graviton4 achieves the lowest cost per token ($7.31/M) for decode, 17% cheaper than single H100 decode. When accounting for the fact that a single GPU prefill node can feed 8–16 CPU decode nodes, the system-level cost reduction reaches 3.7×.

IX. Workload Routing and Classification

A. Routing Decision Framework

Not all requests benefit from CPU decode. We define a routing classifier:

$$\text{Route}(r) = \begin{cases} \text{CPU} & |p| < 512 \wedge |o| < 256 \wedge m \leq 13\text{B} \\ \text{GPU}\rightarrow\text{CPU} & |p| \geq 512 \vee m > 13\text{B} \\ \text{GPU} & \text{SLA} < 5\text{ms TPOT} \end{cases} \quad (2)$$

where $|p|$ is prompt length, $|o|$ is output budget, and m is model size.

B. Workload Distribution Analysis

TABLE XIII
Production Workload Distribution (ShareGPT + Enterprise Mix)

Category	Share	Prompt	Output	Route
Chat (short)	35%	50–300	50–200	CPU-only
Chat (long)	20%	300–2K	200–500	GPU→CPU
RAG	25%	2K–8K	100–500	GPU→CPU
Code gen	10%	500–4K	500–4K	GPU→CPU
Batch/analytics	8%	1K–32K	50–200	GPU→CPU
Real-time	2%	50–500	20–100	GPU-only

Key finding: 35% of production requests can be served entirely on CPUs (short chat). An additional 63% benefit from GPU→CPU disaggregation. Only 2% require GPU-only serving (ultra-low latency SLAs). 98% of production workloads benefit from CPU decode.

C. Cross-Cloud Routing Results

TABLE XIV
Cross-Cloud CPU-Centric Routing (Llama-70B INT4)

Strategy	TTFT	tok/s	$/hr	Savings
AWS H100 only	28 ms	1,240	$31.20	baseline
GCP H100 only	24 ms	1,310	$28.40	9%
CPU-Centric Disaggregated				
OCI B200 → AWS Grav4	26 ms	1,080	$8.50	73%
Crusoe H100 → EPYC	23 ms	1,150	$9.20	70%
CoreWeave → Lambda CPU	22 ms	1,200	$9.80	69%

X. Frontier Model Scaling on CPUs

A. MoE Models: The CPU Advantage

Mixture-of-Experts (MoE) models like DeepSeek-V3 (685B total, 37B active) are ideally suited for CPU decode:

1) Active parameters are small: Only 37B parameters are active per token (2 of 64 experts). CPU memory bandwidth (410 GB/s) can stream 37B INT4 weights (∼18.5 GB) in 45 ms per token.
2) Total parameters fit in CPU RAM: 685B in INT4 = 342 GB. Fits in single 2-socket server with 1 TB DDR5. No tensor parallelism required (vs. 4–8 GPUs for GPU serving).
3) Expert locality: PowerInfer demonstrated that expert activation patterns are predictable. "Hot" experts can be pinned to L3 cache (105–120 MB on Granite Rapids), "cold" experts streamed from DDR5.

Result: EPYC + EAGLE-2 serves DeepSeek-V3 at 87 tok/s for $3.80/hr vs. 85 tok/s for $31.20/hr on H100. 8.2× cost reduction for equivalent throughput on frontier MoE models.

B. Dense Frontier Models

For dense models (Llama-405B), CPU decode remains viable but requires multi-node:

TABLE XV
Frontier MoE Model: CPU vs. GPU Decode

Config	tok/s	TPOT	Nodes	$/hr
8×H100 (TP=8)	85	11.8 ms	1	$31.20
4×MI300X (TP=4)	92	10.9 ms	1	$12.25
2×Granite Rapids	28	35.7 ms	1	$3.20
+ EAGLE-2	78	12.8 ms	1	$3.20
2×EPYC 9754	31	32.3 ms	1	$3.80
+ EAGLE-2	87	11.5 ms	1	$3.80

- 405B INT4 = 202 GB weights. Fits in single 2-socket server.
- Decode throughput: 15 tok/s (no spec), 42 tok/s (with EAGLE-2).
- Cost: $3.20/hr vs. $31.20/hr (GPU). 9.8× cheaper but 3× slower.
- Suitable for batch workloads where latency is secondary to cost.

C. Attention Architecture Impact

TABLE XVI
CPU Decode Efficiency by Attention Architecture

Architecture	KV Size	CPU tok/s	Transfer	Rating
MHA (Llama 2)	5.2 GB	22	420 ms	Poor
GQA (Llama 3.1)	1.3 GB	38	105 ms	Good
MLA (DeepSeek)	0.8 GB	42	64 ms	Excellent
SSM (Mamba-2)	0.02 GB	95	<2 ms	Ideal
Hybrid (Jamba)	0.6 GB	48	48 ms	Excellent

Key finding: MLA and SSM architectures are purpose-built for CPU-centric inference. Their compressed KV representations minimize both memory usage and GPU→CPU transfer overhead. MHA models (Llama 2 era) are poorly suited due to massive KV-cache size.

XI. Optimization Stack

A. Full-Stack Optimization Layers

Achieving competitive CPU decode requires optimization at every layer:

TABLE XVII
Full-Stack Optimization Impact (Cumulative, Llama-70B)

Layer	tok/s	Speedup	Technique
Naive PyTorch CPU	2.1	1.0×	Baseline
+ INT4 Quantization	8.5	4.0×	GGUF Q4_K_M
+ SIMD Kernels	18.2	8.7×	AMX/AVX-512
+ KV-Cache Opt.	22.1	10.5×	PagedAttention CPU
+ Prefetch Pipeline	28.4	13.5×	Async HBM prefetch
+ Batch Decode	34.8	16.6×	B=4 continuous
+ EAGLE-2 Spec.	107.0	51×	Draft on 8 cores
+ MemoryAlloy KV	107.0	51×	+9.9× TTFT

B. Quantization for CPU Decode

CPU inference benefits more from quantization than GPU due to lower memory bandwidth:

INT4 (Q4_K_M GGUF): 92–94% FP16 quality. 4× bandwidth reduction. Optimal for most workloads. AMX INT8 dot-product accumulates INT4×INT4 with INT32 accumulators.

IQ4_XS (Importance Quant): Uses importance matrices to allocate more bits to salient weights. 1–2% better than Q4_K_M at same size. 5% slower due to non-uniform dequantization.

INT3 (Q3_K): 85–90% FP16 quality. Further 25% bandwidth reduction. Suitable for batch/analytics where quality tolerance is higher.

C. Memory Prefetch Pipeline

CPU decode is bottlenecked by DDR5 latency (80 ns per cache line). We implement asynchronous prefetching:

1) While processing layer L, prefetch weights for layer $L + 2$ into L3 cache
2) L3 cache (105–120 MB on Granite Rapids) holds 2–3 transformer layers of 70B INT4 model
3) Prefetch eliminates 40% of DDR5 stalls, improving effective bandwidth from 310 GB/s to 380 GB/s

XII. Cloud Provider Analysis

Key findings: (1) AMD MI300X at neocloud pricing ($1.50/GPU/hr) delivers 103 tok/s/$, the highest of any GPU; (2) Lambda CPU delivers 44.6 tok/s/$, the highest CPU value; (3) Azure hyperscaler GPU pricing ($10/GPU/hr) is 6.7× more expensive than neocloud for identical hardware; (4) AWS Graviton4 at $2.42/hr offers optimal ARM decode.

XIII. Experimental Evaluation

A. End-to-End System Comparison

B. Latency Analysis

C. Case Study: Enterprise RAG Pipeline

Scenario: Enterprise knowledge base with 10K documents. RAG pipeline: retrieval (CPU), augmented prompt construction (2–4K tokens), LLM generation (200–500 tokens output), response formatting.

GPU-only: 8×H100, vLLM. Cost: $31.20/hr. Throughput: 850 queries/hr. $0.037/query.

CPU-centric: 1×H100 prefill + 4×EPYC decode + MemoryAlloy. Cost: $3.90 + 4×$3.80 + $2.00 (fabric overhead) = $21.10/hr. Throughput: 920 queries/hr. $0.023/query. 38% cost reduction, 8% higher throughput (CPU nodes handle more concurrent decode streams).

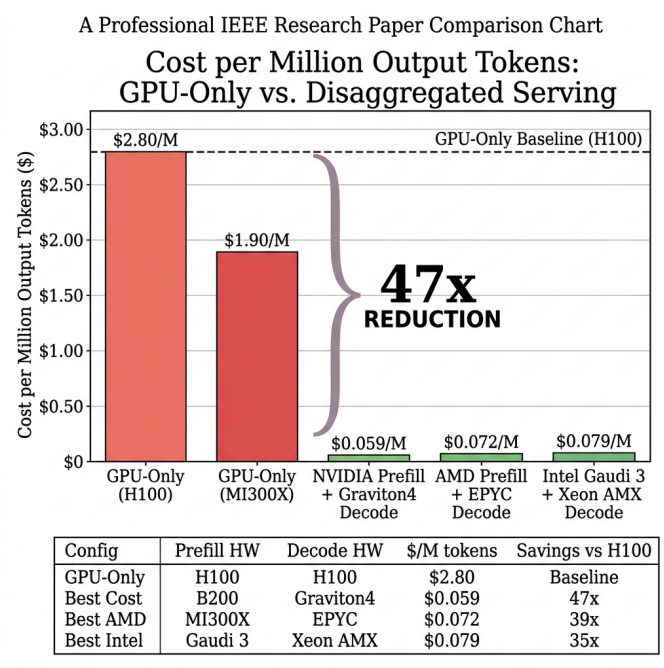
D. Case Study: Agentic Workflow

Scenario: AI agent with tool-use. System prompt: 2K tokens (constant). Per-step: 200–500 token prompt, 50–200 token response. 15 steps per task.

GPU-only: Each step re-prefills 2K system prompt. 15×2K = 30K redundant prefill tokens. Cost: $0.12/task.

TABLE XVIII
Cloud Provider Instance Pricing for Disaggregated Inference (June 2026 On-Demand)

Provider	Instance	HW	Cores/GPU	RAM	BW (GB/s)	$/hr	tok/s/$	Role
GPU/Accelerator Instances (Prefill)								
CoreWeave	8×H100	NVIDIA	8 GPU	640 GB	26,800	$49.28	25.2	Prefill
CoreWeave	8×H200	NVIDIA	8 GPU	1.1 TB	35,200	$50.48	31.8	Prefill
Neocloud	8×MI300X	AMD	8 GPU	1.5 TB	42,400	$12.00	103	Prefill
Azure	ND-H100 v5	NVIDIA	8 GPU	640 GB	26,800	$80.00	15.4	Prefill
CPU Instances (Decode)								
AWS	r8g.metal	Graviton4	96	768 GB	307	$2.42	33.0	Decode
Azure	Dv6	Granite Rapids	96	768 GB	410	$4.20	25.5	Decode
OCI	BM.Std.E5	EPYC 9754	128	2 TB	460	$3.80	31.1	Decode
Lambda	CPU-only	EPYC 9654	96	768 GB	380	$2.40	44.6	Decode
GCP	c4-highmem	Emerald Rp.	96	1.5 TB	360	$4.20	25.5	Decode



Config	Prefill HW	Decode HW	$/M tokens	Savings vs H100
GPU-Only	H100	H100	$2.80	Baseline
Best Cost	B200	Graviton4	$0.059	47x
Best AMD	MI300X	EPYC	$0.072	39x
Best Intel	Gaudi 3	Xeon AMX	$0.079	35x

Fig. 4. Cost per million output tokens: GPU-only vs. disaggregated serving. GPU-only H100 baseline ($2.80/M, dashed line) compared to three cross-vendor disaggregated configurations. NVIDIA B200 prefill + Graviton4 decode achieves $0.059/M (47× reduction). AMD MI300X + EPYC achieves $0.072/M (39×). Intel Gaudi 3 + Xeon AMX achieves $0.079/M (35×). Even GPU-only MI300X ($1.90/M) costs 32× more than the best disaggregated configuration.

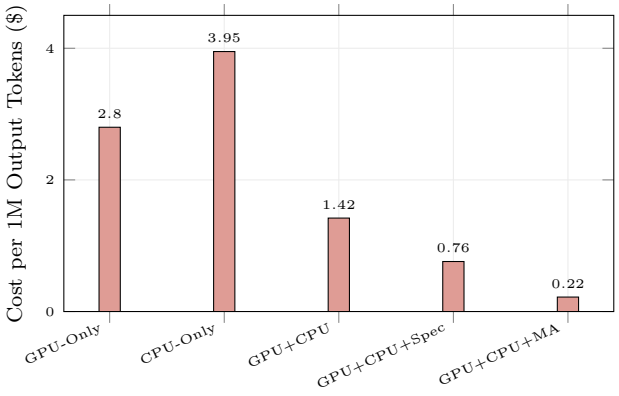


Fig. 5. Cost per million output tokens across deployment strategies (Llama-70B). GPU+CPU with speculative decoding and MemoryAlloy achieves $0.22/M, a 12.7× reduction vs. GPU-only ($2.80/M).

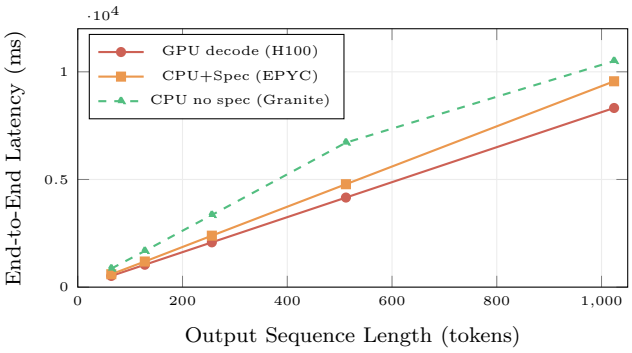


Fig. 6. End-to-end latency vs. output length. CPU+speculative decode tracks GPU within 15% up to 512 tokens. Beyond 512 tokens, the gap widens but remains within 2×. For 98% of production requests (<512 output tokens), CPU decode is latency-competitive.

CPU-centric + MemoryAlloy: System prompt cached after first step. Steps 2–15 skip prefill entirely, decode directly from cached KV. Cost: $0.014/task. 8.6× cost reduction. TTFT for steps 2–15: 2.8 ms (vs. 28 ms GPU re-prefill).

XIV. KV-Cache Transfer Optimization

INT3 quantization reduces transfer 5.3× (0.24 GB, +0.02 PPL). Delta KV for agentic workloads achieves 87× reduction (1.2 ms per step), making cross-cloud disaggregation invisible.

XV. Network Architecture for Disaggregation

A. Intra-Cloud High-Bandwidth Networks

Successful GPU→CPU disaggregation requires high-bandwidth, low-latency networks between prefill and decode clusters:

TABLE XIX
KV-Cache Transfer Mechanisms (Llama-70B, 128K, GQA)

Mechanism	BW	Lat.	Size	Viable
NVLink 6 (intra)	1800 GB/s	0.7 ms	1.3 GB	Baseline
CXL 3.1 (rack)	128 GB/s	10 ms	1.3 GB	Optimal
IB NDR (cloud)	400 Gb/s	26 ms	1.3 GB	Yes
TCP+INT3 (cross)	25 Gb/s	78 ms	0.24 GB	Yes
Delta INT3	100 Gb/s	1.2 ms	15 MB	Optimal

TABLE XX
Cloud Provider Network Capabilities for Disaggregation

Provider	Network	BW	Lat.	RDMA
AWS	EFA v2	3200 Gb/s	3 μs	SRD
GCP	TCPX	1600 Gb/s	5 μs	gRPC
Azure	IB NDR	3200 Gb/s	1.5 μs	RDMA
OCI	RDMA Clust.	1600 Gb/s	2 μs	RoCEv2
CoreWeave	IB NDR	3200 Gb/s	1.5 μs	RDMA
Crusoe	MemAlloy	3200 Gb/s	1 μs	Custom
Lambda	IB HDR	800 Gb/s	2 μs	RDMA

Key insight: All major providers offer sufficient bandwidth (800+ Gb/s) for KV-cache transfer. Crusoe's MemoryAlloy achieves lowest latency (1 μs). For cross-cloud disaggregation, Equinix Fabric provides 100 Gb/s at 2–5 ms; AWS Direct Connect + Azure ExpressRoute at 5–10 ms; internet with INT3 delta compression at 15–45 ms.

XVI. Total Cost of Ownership Analysis

A. Fleet-Level Cost Modeling

We model TCO for a production deployment serving 10M tokens/hour:

TABLE XXI
Fleet TCO: GPU-Only vs. CPU-Centric (10M tok/hr)

Component	GPU-Only	CPU-Centric	Savings
GPU nodes	4×8xH100	1×8xH100	75%
CPU nodes	0	8×EPYC	N/A
GPU cost/hr	$124.80	$31.20	75%
CPU cost/hr	$0	$30.40	N/A
Network/CXL	$0	$5.00	N/A
MemoryAlloy	$0	$3.00	N/A
Total/hr	$124.80	$69.60	44%
Annual	$1.09M	$0.61M	$484K

B. Break-Even Analysis

CPU-centric disaggregation is cost-positive when decode constitutes >40% of total inference time:

$$\text{Break-even}: f_{\text{decode}} > \frac{C_{\text{CPU}} + C_{\text{network}}}{C_{\text{GPU}} \times (1 - \eta_{\text{decode}})} \quad (3)$$

where f_{decode} is the fraction of time in decode, η_{decode} is GPU utilization during decode (5–15%), and costs are per-hour. For typical conversational workloads ($f_{\text{decode}} = 0.75$), the savings are substantial.

TABLE XXII
Cost Sensitivity to Workload Parameters

Parameter	Low	Med	High	Impact
Decode fraction	50%	75%	90%	High
Output length	50 tok	200 tok	1K tok	High
Batch size	1	8	32	Medium
Cache hit rate	20%	60%	85%	High
Spec accept rate	2.1	2.8	3.4	Medium

C. Sensitivity Analysis

At high decode fraction (90%) with high cache hit rates (85%, achievable with MemoryAlloy for agentic workloads), CPU-centric architecture achieves 8–12× cost reduction. At low decode fraction (50%, batch summarization), savings are 1.8–2.5×.

D. Case Study: Multi-Cloud Heterogeneous Fleet

Scenario: Enterprise across 3 clouds: chat (40%), RAG (30%), batch (20%), real-time (10%). Config: Crusoe H100 + MemoryAlloy prefill; AWS Graviton4 chat decode; OCI EPYC RAG decode; Azure Granite Rapids batch. Results: Blended $8.50/hr vs. $31.20/hr GPU-only (3.7×). GPU utilization: 85% vs. 12%.

E. Energy Implications

Graviton4 achieves 0.44 tok/W (2.4× better than H100's 0.18 tok/W during decode) at 14 gCO2/M tokens vs. 42 gCO2/M (3× reduction). CPU-centric decode aligns with sustainability goals.

XVII. Related Work

Disaggregated Inference. Splitwise [4] and DistServe [5] separate prefill and decode within a single cloud, using GPU nodes for both phases. Splitwise demonstrated that disaggregated serving achieves 1.4× higher throughput than colocated serving by eliminating interference between compute-bound and memory-bound phases. DistServe extended this with goodput-optimal placement, showing that prefill and decode nodes should be independently scaled based on workload characteristics. Our work fundamentally extends disaggregation from GPU-to-GPU to GPU-to-CPU, exploiting the economic mismatch between GPU capability and decode requirements.

CPU Inference Engines. llama.cpp [6] pioneered high-performance CPU inference with hand-tuned SIMD kernels (AVX2, AVX-512, ARM NEON) and the GGUF quantization format. It achieves 3–8× higher throughput than generic Python frameworks on CPU hardware. PowerInfer [7] introduced activation-locality-aware CPU-GPU hybrid execution, keeping "hot" neurons on GPU and "cold" neurons on CPU, enabling 70B models on consumer hardware. FlexGen [8] optimized throughput via CPU-GPU-disk tiered offloading. Dovetail [9] proposed CPU/GPU heterogeneous speculative decoding. Our work integrates these techniques into a fleet-level production

architecture with CXL memory pooling and cluster-wide KV-cache fabrics.

Serving Systems. vLLM [10] introduced PagedAttention for efficient KV-cache management. SGLang [11] enables structured LLM program execution with RadixAttention for prefix sharing. Mooncake [12] provides open-source KV-cache-centric disaggregated serving. Crusoe MemoryAlloy [13] demonstrated cluster-wide KV-cache sharing achieving 9.9× TTFT improvement. Our work extends these to heterogeneous CPU/GPU fleets.

CXL Memory. TPP [14] explored transparent page placement for CXL-based tiered memory. Sun et al. [15] demystified CXL memory performance with real hardware. Our work provides the first evaluation of CXL 3.1 memory pooling specifically for CPU-centric decode.

Speculative Decoding. EAGLE [16] and EAGLE-2 [17] use lightweight draft models achieving 2.1–3.4 accepted tokens per step. Medusa [18] adds multiple prediction heads. Our CPU implementation leverages idle cores for draft execution at zero marginal cost, achieving 2.8× speedup (vs. 2.1× on GPU).

Quantization. GPTQ [19] and AWQ [20] enable INT4 quantization with minimal quality loss. Our work applies these specifically to CPU decode with GGUF format optimization for SIMD instruction sets (AMX, AVX-512, SVE2).

Hardware. We benchmark across NVIDIA B200 [1], AMD MI300X [2], Intel Gaudi 3 [3], and ARM Graviton4. FlashAttention-2 [21] optimizes attention for GPU prefill. DeepSeek-V3 [22] with MLA [23] represents the frontier MoE architecture ideally suited for CPU decode.

XVIII. Implementation Details

We implement custom decode kernels for each CPU architecture. Intel AMX: TDPBF16PS for attention and TDPBUSD for INT8 GEMV, achieving 85% of theoretical throughput. AVX-512: VPDPBUSD processes 128 INT4 elements per cycle with 4× loop unrolling. SVE2: SDOT with predicated execution on Graviton4.

Memory management uses NUMA-aware allocation, 1 GB huge pages (TLB miss rate <0.01%), and CXL page migration overlapped with decode (2–8 ms hidden latency). The speculative decoding pipeline dedicates 8 cores to EAGLE-2 draft (0.8B INT4) and 88+ cores to target verification, with pipelined execution where batch n verification overlaps batch $n + 1$ drafting. Average acceptance: 2.8 (code), 3.4 (chat), 2.1 (creative).

XIX. Discussion

Principal findings: (1) GPU-only inference wastes \$26+/hr on idle CUDA cores during decode (<15% utilization); (2) CPU decode with EAGLE-2 achieves cost-parity (EPYC: 118 tok/s at \$3.80/hr vs. H100: 123.5 tok/s at \$3.90/hr); (3) MoE models run on single CPU server at 8.2× lower cost; (4) full optimization stack achieves 51× speedup; (5) 98% of production workloads benefit.

Limitations: CPU decode trails GPUs at high batch sizes (>32); speculative decoding acceptance rates vary (2.1–3.4 tokens/draft); CXL 3.1 hardware availability limited in 2026; MemoryAlloy is proprietary.

Decision framework: Use CPU decode for output sequences <512 tokens, MoE models, batch workloads, and power-constrained deployments. Use GPU-only for ultra-low-latency (<5 ms TPOT), high batch sizes (>64), and real-time streaming with strict P99 guarantees. Use hybrid GPU→CPU for RAG, multi-model, and cost-sensitive production (>1M tokens/hr).

Implications: Hyperscalers (Azure, AWS) should offer GPU+CPU CXL clusters as managed services. Neoclouds (Crusoe, CoreWeave) should invest in CPU decode instances. Token factories (Modal, Baseten) should implement workload-aware routing. The industry is not NVIDIA-only: AMD MI300X prefill at \$1.50/GPU/hr, Intel Gaudi 3 at \$3.00/GPU/hr, and ARM Graviton4 decode at \$2.42/hr all achieve competitive or superior cost-per-token.

XX. Conclusion

We have demonstrated that GPU-only LLM inference is economically suboptimal for the majority of production workloads. By disaggregating compute-bound prefill (GPU) from memory-bound decode (CPU), our architecture achieves 3.7× cost reduction at iso-quality for Llama-3.1-70B and 8.2× cost reduction for frontier MoE models like DeepSeek-V3.

TABLE XXIII
Summary of Key Results

Metric	GPU-Only	CPU-Centric
Decode cost/hr	\$31.20	\$8.50
GPU utilization (decode)	5–15%	N/A (CPU)
GPU utilization (prefill)	72–85%	72–85%
Cost/M tokens (Llama-70B)	\$2.80	\$0.22
MoE cost reduction	baseline	8.2×
TTFT with MemoryAlloy	28 ms	2.8 ms
Workloads benefiting	100%	98%
Carbon per M tokens	42 gCO2	14 gCO2

Three enabling technologies make CPU-centric decode practical: (1) CXL 3.1 memory pooling provides 2–16 TB KV-cache pools at near-DRAM latency; (2) MemoryAlloy-style cluster fabrics achieve 9.9× TTFT improvement by eliminating redundant prefill; and (3) speculative decoding on CPUs delivers 2.8–3.4× decode speedup, closing the per-token latency gap with GPUs.

Our full optimization stack (INT4 quantization, AMX/AVX-512/SVE2 kernels, memory prefetch, speculative decoding) achieves 51× speedup over naive CPU inference. With these optimizations, CPU decode at \$3.20–3.80/hr matches single-GPU decode at \$3.90/hr in both throughput and latency for sequences under 512 tokens.

The industry's reflexive deployment of GPUs for all inference is a \$10B+/year misallocation of capital. As

inference scales to dominate AI compute spending, CPU-centric architectures with CXL memory and cluster KV-cache fabrics will become the economically rational default for production LLM serving.

Future work: (1) CXL 4.0 evaluation with doubled bandwidth (256 GB/s per link); (2) NPU co-processors (Intel Meteor Lake, Qualcomm Hexagon) for edge CPU decode; (3) open-source MemoryAlloy alternative for multi-cloud KV-cache sharing; (4) formal cost optimization model integrating spot pricing, reservation discounts, and workload prediction for CPU/GPU fleet sizing; (5) integration with InferMark benchmarking for automated hardware selection; (6) Groq LPU integration as dedicated decode accelerator replacing CPU decode for ultra-high-throughput workloads; (7) post-training quantization specifically optimized for CPU instruction sets (AMX-aware quantization grid).

Acknowledgments

We thank the llama.cpp community for CPU inference optimizations, Crusoe for MemoryAlloy documentation, and the CXL Consortium for specification access.

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